AMPHORA / BUSINESS PLANNING / 14 OCTOBER 2026

Early-stage compute estimates

First three months live • two simultaneous campaigns • 20 ML node releases per week

Recommended planning budget: $6,500/month; roughly $19,500 for 13 weeks.
The reference bill is $6,327/month including a 20% reserve. This funds a CPU-first platform with managed state, separate campaign capacity, elastic training, API-backed agents and modest GPU experimentation. It is an estimate, not a measured service-level commitment.

Recommended starting decision

Use **one primary cloud and one region** for authoritative state, artifacts, campaign execution and the inherited-model training queue. Start on AWS if there is no existing commercial or technical preference; Azure is an equally credible choice and is modestly cheaper in the selected CPU examples. Use GCP when an existing data/ML workflow makes it the easier platform to operate. Decide before launch and keep that choice stable for 90 days.

The inspected Amphora property paths are mainly RDKit features with boosting, ridge and physical baselines. The separate NOMAD CrystalAtlas program in v1 also has GPU-backed tree models and a large feature bank. Keep a GPU burst path available; benchmark that mix before committing to a permanent fleet. A specialist supplier can host workers while authoritative state stays in the primary cloud. [L05–L11]

Planning case	All-in / month	Meaning
Lean	$2,594	Modest datasets and tightly bounded tuning; reduced recovery posture.
Base	$6,327	Larger classical-ML datasets, 10 trials × 5 folds; separate live capacity.
Stress	$32,267	Large dense feature matrices, broad tuning and higher agent/data activity.

Included: hosting, training/inference compute, persistent storage, backups, network allowances, logs/secrets/registry, staging and model/API usage. **Excluded and shown separately:** engineering/scientific labor, commercial data licenses, wet-lab work, taxes and negotiated support contracts. No credits, spot discounts or annual commitments are needed to make the base case fit.

Critical qualification: the current demo supports one active campaign. Two concurrent campaigns, cloud deployment, operational agents and the 20-release training factory require software and deployment acceptance work. The hardware estimate does not establish those capabilities. [L01–L04]

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What the existing footprint tells us

Area	Observed today	Planning implication
Owner services	Six scientific/state owners plus browser gateway; Deployment supervises the installation.	Co-location can be economical, but service packages, roles and state ownership stay isolated.
Durable state	PostgreSQL, Qdrant and immutable scientific artifacts; versioned receipts and retained history.	Storage, backup consistency and restore testing are launch costs, not later extras.
Scientific tools	Independently installed Hansen/Arrhenius tools, plus supplied-structure qualification.	Scale independently packaged workers; keep expensive jobs out of API processes.
Campaign admission	One active campaign in the accepted local profile. Bounded resource admission is still proposed.	Two simultaneous campaigns need explicit scheduling, admission and fairness changes.
Deployment	Windows x86_64 local profile; independently installed owners. Broad Linux/cloud/container hosting not accepted.	Budget a cloud/OS qualification effort. Existing package isolation helps, but is not a ready cloud installer.
ML and agents	Inherited scientific model integration is queued; proposed operational agents are not built.	Model recovery, reproducible training, agent budgets and remote worker grants remain delivery work.

A deployment recommendation, not a redesign

Keep Runtime as execution authority. A batch scheduler allocates resources to a reviewed training/tool adapter; it does not take over campaign semantics, scientific acceptance or provenance. Remote workers receive scoped input/artifact access and return outputs through owned contracts. Avoid broad peer database access or a catch-all cloud backend. [L01–L04]

A managed PostgreSQL installation may serve separate owner schemas and roles. Reconcile its supported extensions, migration privileges, transaction/fencing behavior and backup restoration before accepting it. Qdrant is a rebuildable retrieval projection; it must not become the only copy of the scientific corpus.

Before two campaigns can go live: qualify concurrent admission, run ownership, per-campaign quotas, shared resource fairness, stale-worker rejection, recovery with both campaigns active, and worker/version replacement. More CPUs alone cannot remove the current single-campaign contract.

Evidence basis: repository checkpoint 75eaa03b, local architecture and active trackers reviewed read-only on 2026-10-14. No live performance benchmark or cloud provisioning was run for this document. The synthetic demo is not a chemistry throughput benchmark.

90-DAY INFRASTRUCTURE PLAN

The v1 models are the right starting reference

Reference family	Workload evidence	Compute reading
BigSolDB, later signed benchmark [L05]	94,532 modeled rows; 433 features; RDKit 2D descriptors + LightGBM; script uses 8 threads. Two recorded primary-protocol runs: 271.6 and 247.7 seconds.	CPU-friendly empirical anchor: about 4.1–4.5 minutes for that recorded protocol. Not a timing for every split, tuning search or new host.
Other classical paths [L06, L11]	Earlier BigSolDB: 74,287 rows / 290 features. Polyomics: 78,083 polymers / 528 features. Other paths include RandomForest, ridge and logistic regression.	CPU-first fits and feature work. Benchmark each family; do not equate historical training versions.
Polyverse melt viscosity [L07]	1,380 measurements; 280 features; boosting / ridge comparisons.	The reviewed inherited size is small. Larger polymer corpora are a growth assumption.
Polyverse solvent diffusivity [L08]	3,044 measurements; 548 features; source-held-out evaluation.	Validation design matters more than buying an accelerator for the receipt-sized dataset.
Purformance coating properties [L09]	65 formulations across four targets in one receipt; 86 in scratch-hardness receipt.	Small-data fit cost; quality, deduplication and applicability dominate release effort.
Separate CrystalAtlas program [L10]	At least 8,000 registered descriptors; CatBoost GPU, XGBoost CUDA and LightGBM GPU helpers. External corpus not inspected.	Already GPU-oriented code, not a hypothetical GNN. Separate from Amphora; include only if selected for the weekly mix.

These are historical workloads, not newly reproduced timings or qualified v2 releases. The 4-minute BigSolDB record covers a bounded primary-protocol replay, not the full split/tuning program. v1/README.md identifies NOMAD as a separate program; its hardware and operational history are not Amphora deployment evidence. [L05, L10]

Why the recommendation is CPU-first

For Amphora's tabular property models, cache features by structure and feature version. Control LightGBM/OpenMP/BLAS threads per job: many fits each requesting 28 threads can oversubscribe a machine. CrystalAtlas-like 8,000-feature matrices need a separate memory benchmark: 1m rows × 8,000 float64 features = 64 GB before copies. [S30]

For CPU-first paths, qualify GPU backend changes against the accepted validation protocol. Existing GPU-coded paths require compatible drivers/frameworks and cannot be assumed to fall back to CPU. The base budget's 150 GPU-hours are a small initial allowance, not a measured CrystalAtlas requirement. [S31]

90-DAY INFRASTRUCTURE PLAN

Turn “20 nodes per week” into a capacity model

For budgeting, a **node release** means one independently packaged, versioned trained-model bundle with its feature code, dataset/split identity, validation evidence and inference adapter. A campaign occurrence is an invocation of a node; a new replica is not a new model. Twenty releases per week means approximately **86.7 releases/month or 260 over 13 weeks**.

Assumption to settle before procurement: the release target is not 20 newly invented scientific capabilities every week. Compute can train and evaluate bundles; it cannot guarantee new endpoints, lawful labeled datasets, useful predictive performance or scientific acceptance at that rate. Those are separate people/data constraints.

Scenario	Dataset growth envelope*	Training program per release
Lean	Up to ~250k rows, ~550 features; cached 2D descriptors / fingerprints.	5 trial configurations × 5 folds + 1 final fit; 0.10h/fit + 0.5h preprocessing.
Base	~1m rows, ~550 features; higher validation and packaging volume.	10 trials × 5 folds + 1 final fit; 0.25h/fit + 1.5h preprocessing.
Stress	~5m rows, up to 2,048 dense features; larger memory class.	30 trials × 5 folds + 1 final fit; 0.50h/fit + 4h preprocessing.

*These are planning envelopes, not observed v1 dataset sizes. Row count is not a timing predictor by itself: number of unique molecules, feature cost, sparsity, targets, tree depth and validation grouping all matter. Timings are deliberately conservative assumptions; no claim of linear scaling from the historical BigSolDB run is made.

What each training hour includes

A worker-hour is an elapsed hour of the specified complete VM while allocated to a job, not one CPU-core-hour. The lean/base worker is 16 vCPU / 128 GiB; stress is 32 vCPU / 256 GiB. Folds/trials may run on separate workers, but their aggregate billed hours still count. Another 20% covers extra baseline/calibration checks, repeats, failures and provisioning overhead.

Multiple outputs, seeds, alternative feature sets or repeated calibration multiply the fit count. A two-seed program roughly doubles the affected fits; a fresh dataset per node increases preprocessing and data movement. Grouped/scaffold/polymer-held-out protocols must survive optimization; cheaper leakage-prone validation is not an acceptable capacity shortcut.

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Training capacity: elastic slots, not permanent GPUs

Metric	Lean	Base	Stress
Fits / release	26	51	151
Elapsed hours / release, before 20% overhead	3.10	14.25	79.50
Billed worker-hours / week	74	342	1,908
Billed worker-hours / month	322	1,482	8,268
On-demand price / complete VM-hour	$1.0584	$1.0584	$2.1168
CPU training cost / month	$341	$1,569	$17,502
Concurrent slots at 65% of 168h/week	1	4	18

Prices are current AWS US East Linux r7i.4xlarge / r7i.8xlarge examples. Memory-class prices include the whole VM, not only the CPU. Disk, transfer and artifacts are budgeted elsewhere. [S01]

Base lane: allow four 16-vCPU / 128-GiB training slots, with a separately budgeted burst ceiling of eight after quota and deadline checks. Four slots have 436.8 worker-hours/week at the 65% planning utilization, versus 342 required. Two slots have only 218.4 hours/week and will not sustain the assumed base program.

The 65% factor is capacity headroom, not a billed discount. Jobs should release their workers when complete; the price model includes actual allocated hours plus the stated overhead. Keeping four VMs on continuously would cost about $3,091/month before disks, versus $1,569 in the base elastic training lane.

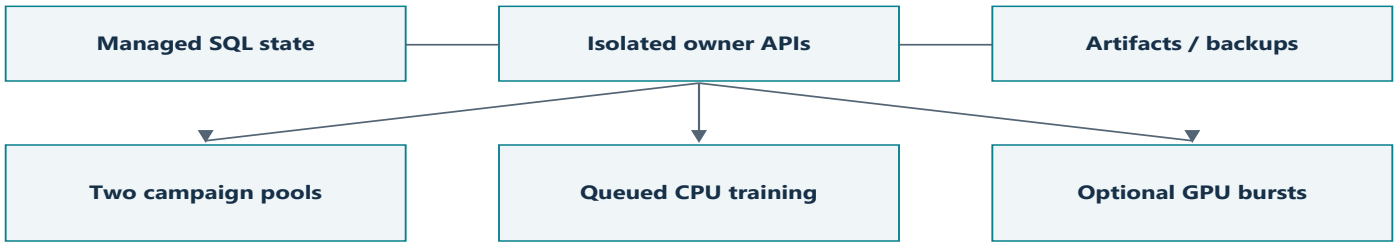
The weekly deadline changes the required ceiling

If all 20 releases must finish in five days, the base case needs five slots at the same utilization allowance. A 48-hour weekly training window needs eleven. This is why a monthly GPU/server bill alone cannot establish throughput. Trial/fold tasks must be distributable for these slot calculations to hold; measure the longest unsplittable job and cold-start/data staging time.

Request training quota for 64 vCPUs initially and a pre-agreed 128-vCPU burst path, in addition to operational/campaign capacity. Quota approval and a price listing do not guarantee stock in the selected zone.

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Two full campaigns need protected live capacity



Start with an 8-vCPU / 32-GiB control host, two separate 8-vCPU / 32-GiB campaign worker allocations, and separate retrieval and agent allocations. Keep training contention away from coordination and inference. These are initial benchmark targets, not accepted campaign capacity figures.

Lane	Initial allocation	Scale rule
Owner APIs / coordinator	1 × 8 vCPU / 32 GiB; separate packages and roles.	Increase only when coordination latency, database contention or projection lag warrants it.
Campaign A and B	2 × 8 vCPU / 32 GiB total; per-campaign limits.	Start with 2–4 concurrent lightweight invokes each; cap threads and memory per invocation.
Training	0–4 × 16 vCPU / 128 GiB; queued / elastic.	Does not consume the protected campaign worker allocation.
Agents / retrieval	4 vCPU / 16 GiB each, separate execution ownership.	Index rebuilding and LLM calls have their own quotas; scale retrieval with measured index size.

Campaign count is not enough to predict workload

Illustration: two campaigns × 10,000 candidates × 10 evaluated model nodes × 1 CPU-second/invocation = **55.6 vCPU-hours** for the pair, before orchestration and feature overhead. Twenty such pairs/month require about 1,111 vCPU-hours. The two proposed 8-vCPU pools provide about 7,592 vCPU-hours/month at 65% utilization, if work parallelizes well.

At 10 CPU-seconds per invocation that illustration becomes 11,111 vCPU-hours and exceeds the same pools. Expensive conformer search, simulation, geometry-dependent features or synchronous per-candidate LLM reasoning could change the bill much more. This illustration is not an assumed campaign volume or a measured throughput claim.

Accept two representative full campaigns while a base training burst and agents are active. Measure p95 invoke latency, queue age, total CPU-seconds, peak memory, API latency and recovery—not simply whether two runs start.

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Storage and data movement can erase compute savings

Storage layer	Initial planning envelope	Treatment
Authoritative artifacts	~3 TB live object data: source revisions, cached features, released models and campaign results.	Immutable versioned keys; deduplicate by content identity. Preserve rights and lineage.
Independent recovery copy	~3 TB retained copy / incremental versions, plus SQL backup policy.	Budgeted in the $180 object allowance; reprice retention and regional replication before deployment.
Hot block / scratch	~1 TB across control/retrieval and active-worker scratch; $200 allowance.	Temporary job disks scale down with workers. Do not leave every trial disk indefinitely.
Model releases	Illustration: 260 bundles × 5 GB = 1.3 TB over 13 weeks.	5 GB is an allocation assumption; classical models may be far smaller. Persist accepted bundles, not every failed trial.

RAM is different from raw dataset storage

A dense float64 matrix is rows × features × 8 bytes: 1m × 550 = **4.4 GB**; 5m × 2,048 = **81.9 GB**, before pandas/object overhead, folds, bins, models and copies. That stress matrix can exhaust a 128-GiB worker even though the raw file fits on disk. Five million packed 2,048-bit fingerprints are only 1.28 GB; unpacking them into float64 changes the footprint by 64×.

Use cached immutable features, columnar files and streaming/bounded loading where the model permits. Cache by unique molecular identity and feature version, not by every assay row. Retrieval memory also scales separately: 1m × 768 float32 embeddings are 3.1 GB before index structures and replicas; measure actual Qdrant overhead.

Cross-provider training: move once, reuse many times

At an illustrative $0.09/GB outbound rate, moving 1 TB once costs about $90. Moving 250 GB for every one of 86.7 monthly jobs costs about $1,950. NAT processing at $0.045/GB can add another $975 if that path is used. Confirm actual region/tier charges; these examples omit free allowances conservatively. [S02] [S28]

Keep CPU training close to canonical data. For a specialist GPU lane, cache an approved dataset revision and return compact results. Use AWS S3 gateway endpoints where appropriate to avoid routing artifact traffic through a charged NAT gateway. The external supplier may charge its own persistent disks; “no egress fees” there does not waive the primary cloud’s outbound charge. [S29] [S10]

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Reference monthly bill: approximately $6,500

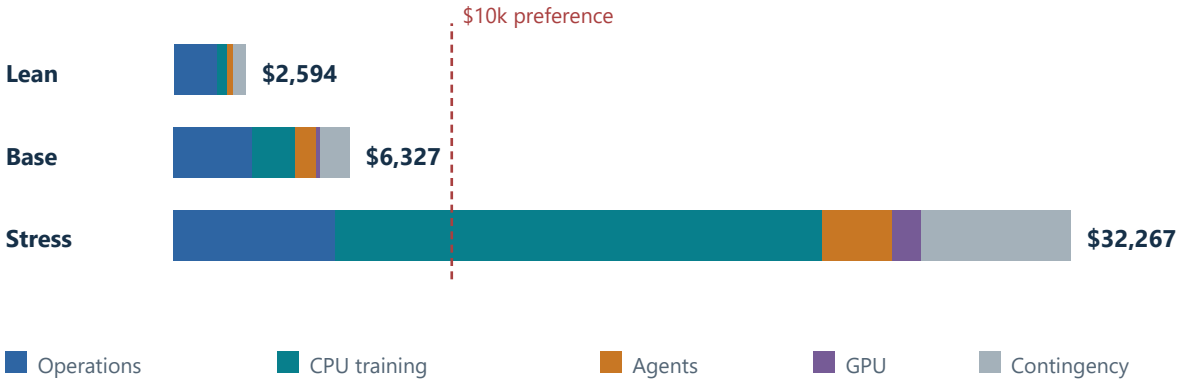
Line item	Basis	USD / month
Control / owner APIs	1 × m7i.2xlarge × 730h	$294
Two campaign worker pools	2 × m7i.2xlarge × 730h	$589
Qdrant / retrieval VM	1 × m7i.xlarge × 730h	$147
Operational agent runner VM	1 × m7i.xlarge × 730h	$147
Managed PostgreSQL + standby	Planning allowance; ~4-vCPU / 16-GiB primary class + standby; storage included	$750
Block disks / training scratch	Planning allowance across steady hosts and temporary jobs	$200
Object artifacts + backup copy	Planning allowance for ~6 TB total retained bytes + requests	$180
Networking / ingress / egress	Planning allowance; no repeated bulk cross-cloud transfers	$150
Logs, secrets, registry, support reserve	Planning allowance; negotiated support extra if required	$200
Staging / restore drills	Planning allowance; bounded test installations, cleaned up	$150
Elastic CPU training	1,482 worker-hours × $1.0584; overhead already included	$1,569
Optional GPU burst allowance	150h × g6.2xlarge $0.9776; no permanent GPU required	$147
All campaign + operational agent APIs	Shared allowance; token model on page 14	$750
Subtotal	Exact unrounded calculation; displayed rows rounded	$5,273
Contingency	20% of subtotal; separate from training overhead	$1,055
TOTAL / MONTH	Planning provision; cash spend can be lower	$6,327

Four steady 8-vCPU-equivalent VM allocations total $1,177.34/month at verified AWS Linux US East prices. Database, disks, retention, networking and operations are explicit allowances, not obtained vendor quotes. A managed database standby improves database availability; it does not make the single control host highly available. [S01] [S32]

Planning range: **$5.5–7.5k/month** around the base assumptions; reserve access up to $10k for agreed bursts. Budget ~$19.5k for 13 weeks, plus a separate setup cash allowance of ~$300–1,000 for data import and deployment artifacts, excluding labor. No purchase, paid model request or provisioned cloud account is implied by this report.

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How the budget changes with the workload



Monthly line	Lean	Base	Stress
Always-on platform / storage / operations	$1,571	$2,807	$5,813
CPU training	$341	$1,569	$17,502
Campaign + operational agent APIs	$250	$750	$2,500
Optional GPU work	$0	$147	$1,074
20% contingency	$432	$1,055	$5,378
Total / month	$2,594	$6,327	$32,267
13-week / 3-month planning total	$7,783	$18,981	$96,800

Lean: two 4-vCPU campaign pools, no separate agent VM, single-AZ managed database and smaller storage/monitoring allowances. It can be enough for receipt-sized or moderately larger v1 workloads after measurement. It is not the same recovery or live-load posture as the recommended base.

Stress: 32-vCPU / 256-GiB training workers, 30 trials per release, larger live pools, doubled database allowance, and more storage/network/agent use. CPU training alone reaches about $17.5k/month. The stress case is not an inevitable year-one cost; it is a concrete warning that broad tuning on dense large data breaks the preferred budget.

$1k/month is a constrained pilot budget, not the recommended provision for this requirement. It may work with rented CPU hardware, v1-sized jobs, low agent activity and reduced recovery guarantees, but should not be represented as enough for the assumed base training program plus two continuously protected campaigns.

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AWS, Azure and GCP: compare whole workloads

Provider / region basis	CPU examples, Linux on-demand	GPU examples, complete VM
AWS US East (N. Virginia)	8 vCPU / 32 GiB m7i.2xlarge: $0.4032/h. 16 / 128 r7i.4xlarge: $1.0584/h. [S01]	g6.xlarge, 1 L4 / 24-GB nominal VRAM, 4 vCPU / 16 GiB: $0.8048/h. g6.2xlarge, 8 / 32: $0.9776/h.
Azure East US	D8as_v5, 8 / 32: $0.3440/h. E16as_v5, 16 / 128: $0.9040/h. [S03] [S04]	NC4as_T4_v3, 1 T4 / 16-GB VRAM: $0.526/h. NC24ads_A100_v4, A100 class: $3.673/h.
GCP public US / Iowa reference	n2-standard-8, 8 / 32: $0.388472/h. n2-highmem-16, 16 / 128: $1.048112/h. [S05]	g2-standard-4, 1 L4 / 24-GB VRAM, 4 / 16: $0.706832/h. a2-highgpu-1g, 1 A100 / 40-GB VRAM: $3.673385/h. [S06]

Choice	Advantages for Amphora	Costs / tradeoffs
AWS	One home for relational state, artifacts and elastic CPU/GPU jobs; AWS Batch can allocate workers without a separate scheduler service fee.	IAM/network setup, logs, NAT and transfers require ownership. Prefer existing public contracts over a proprietary ML-only control path.
Azure	Competitive selected CPU rates; a reasonable fit if Microsoft identity, contracting or Windows operations are already preferred. Azure Batch supports the burst lane.	DPAPI/local installation does not port automatically. Windows licensing and managed-service compatibility must be quoted and qualified.
GCP	Competitive L4 examples and flexible CPU shapes; useful when dataset/ML workflows already live there. Google Batch is available.	The cheapest GPU-only listing is not the whole VM price. Quotas, regional stock and data placement still matter.

AWS, Azure and GCP Batch each charge for underlying resources rather than adding a Batch scheduler service fee. They do not remove disk, data transfer or optional managed ML platform charges. [S19] [S20] [S21]

Holding every non-VM allowance and the $147 GPU allowance constant, the selected base shape projects about **$6,327 AWS, $5,845 Azure and $6,257 GCP**/month including reserve. This isolates CPU list-price differences; it is not three obtained full-stack quotes or a performance-normalized benchmark. GCP figures use the fetched public US table; confirm the selected region/SKU before purchase.

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Specialist ML clouds: useful burst suppliers

Provider	Observed public price anchor	Best use / principal tradeoff
Runpod	Pod listing: L40S 48 GB $1.09/h; A100 80 GB $1.79/h; H100 PCIe 80 GB $2.89/h. [S07]	Short CUDA jobs / inference experiments. Confirm tier, region, RAM and stock at launch; the public page has tier controls. Use qualified Secure Cloud terms for proprietary work. [S08]
Lambda	Single GPU: A6000 48 GB $1.09/h; A100 40 GB $1.99/h; H100 PCIe $3.29/h, SXM $4.29/h. [S09]	Straightforward VM training with substantial included CPU/RAM. First-come availability; persistent filesystems are separately billable. [S10]
Hyperstack	A6000 48 GB $0.50/h; L40 48 GB $1.00/h; A100 80 GB $1.35/h; H100 PCIe $2.50/h. [S11]	Attractive published GPU economics. Check the full chosen flavor, eligible location and storage bill; do not infer equal performance from the GPU label.
Paperspace / DigitalOcean	A100 advertised $1.15/h; product plan / multi-GPU eligibility applies. [S13]	Convenient development workflows. The headline has conditions; verify minimum bundle and powered-off storage charges before budgeting.
CoreWeave	A100 80-GB 8-GPU node: $21.60/node-hour; $2.70/GPU-hour equivalent. [S12]	Strong option for larger sustained GPU workloads. Training may require paying the entire node, not a single $2.70 GPU. Confirm access/contract terms.
Vast.ai	Live host offers; no single guaranteed rate used here. [S14]	Exploratory public-data jobs with portable checkpoints. Marketplace host variability makes it a weaker default for sensitive, deadline-critical work.

This is a shortlist, not a finding that one vendor is most reliable or fastest. Compare a benchmarked **cost per accepted release**, queue availability, upload time, checkpoint storage and operational effort. L40, L40S, A6000, A100 and H100 are different devices; posted dollar/hour figures cannot be ranked as identical throughput.

Recommendation: choose at most one specialist supplier as a pre-qualified optional GPU lane. Keep canonical datasets, scientific identity and receipts in the primary deployment. Do not introduce multi-cloud control, databases or a general vendor-routing platform in the first 90 days.

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When GPUs actually change the requirement

Work	Initial placement	Upgrade trigger
RDKit 2D features, ridge, scikit-learn boosting	CPU workers; feature cache and controlled parallelism.	Measured CPU bottleneck after correcting redundant featurization and oversubscription.
Existing CrystalAtlas GPU tree paths [L10]	Single-GPU benchmark; measure VRAM, host RAM, features and CUDA/OpenCL dependencies.	If selected for the weekly mix, GPU training is required; qualify before launch. Separate program is not yet an Amphora node.
Graph neural nets / learned molecular embeddings	Start with one 24–48-GB GPU; use 80 GB if the measured working set needs it.	Accepted new model family; batch/graph size and memory measured. Not the base v1 reference.
Large molecular / foundation models	Explicit separate training program and budget.	A model/dataset requiring multi-GPU memory, distributed training or long GPU-hours.
Molecular dynamics / quantum / heavy simulation	Separate HPC/scientific capability lane.	Selected simulator, system size, wall-time and software license; no implied coverage here.

Illustrative GPU expansion: 20 releases/week × 20 aggregate GPU-hours/release × 52/12 × 1.25 overhead = **2,167 GPU-hours/month**. At a posted $1.79/h it costs ~$3,878; at a $3.673/h complete A100 VM rate it costs ~$7,958 before data, preprocessing and the always-on platform. Those devices have different memory/CPU configurations; this is a rate sensitivity, not an equal-speed comparison. [S07] [S04]

The same GPU-hours require roughly four GPU slots at 65% of 168 hours/week. A single 24/7 GPU cannot deliver that volume. If GPU models replace CPU releases, remove those CPU fits from the bill; if they are additional, add the cost. The 150h base GPU allowance must be reset from the selected CrystalAtlas-like workload. Verify bundle minimums before committing.

Spot is a savings path, not the base feasibility argument

Spot can be interrupted: AWS generally provides two minutes' notice (with exceptions), Azure documents 30 seconds and no Spot SLA, and Google documents 30-second reclamation notice. Treat notices as best effort; checkpoint proactively. [S22] [S23] [S24]

Use spot only for restartable fits with immutable inputs, safe output publication, retries and a paid on-demand fallback. Keep authoritative services and required live-campaign baseline off spot. After qualification, trial spot on a bounded fraction of training; do not promise an unmeasured discount or deadline.

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Rented hardware: lower fixed cost, more ownership

Rental form	Advantages	Tradeoffs for the first 90 days
Vendor-hosted dedicated CPU servers	Predictable monthly pricing; strong CPU/RAM per dollar; local NVMe and stable cache. No purchase depreciation.	Full OS/database/backup responsibility unless managed support is bought. Less elastic resizing; replacement latency and spare capacity need contractual answers.
Vendor-hosted GPU bare metal	Guaranteed device allocation when contracted; local dataset reuse; potentially economical sustained GPU occupancy.	Pay when idle. Exact GPU count, RAM, setup charges, stock and support change the economics.
Physical equipment delivered to the company	Private local data, familiar control, possible reuse of existing space and expertise.	Add shipping/setup, power/cooling, UPS, networking, remote access, physical protection and repair/swap terms. Local internet/power remains a failure domain.

Examples: Hetzner AX102 publicly specifies 128-GB ECC RAM and two 1.92-TB NVMe drives in European locations; that machine is not four 128-GiB training slots. OVHcloud has US CPU bare-metal catalogs and dedicated GPU options. A posted GPU example, Scale-GPU-1, starts at $1,145/month plus $1,145 setup and is marked “soon available”; GPU count/location require a quote. HGR-AI-2 starts at $4,355/month plus equal setup. [S16] [S17] [S15]

For delivered equipment, obtain a written 90-day rental quote from an equipment vendor such as Rentacomputer; its server page offers GPU sourcing by quotation. No procurement or vendor outreach has been performed. [S18]

A viable CPU-rental alternative

Ask for a small control/retrieval host plus **two training servers, each with at least 16 physical cores / 32 threads, 384–512 GB ECC RAM and mirrored NVMe**. Together they can expose four 16-thread / 128-GB reservations with RAM left for the OS, subject to benchmarking. Physical cores, SMT threads and cloud vCPUs are not performance-equivalent units.

With a hypothetical $900/month combined training-host quote, $100/month amortized setup and $150/month extra maintenance reserve, the worker-lane cost is $1,150 versus the base AWS elastic $1,569. Break-even is ~1,087 billed AWS worker-hours/month at equal benchmarked throughput. The benefit is ~$419/month, not automatically thousands. A full dedicated-platform alternative may cost ~$3.5–5.5k/month including agents/backups/reserve, but that range is quote-dependent and is not the reference bill.

GPU rental needs sustained occupancy

Illustration, not a vendor quote: $500/month for one GPU + $300 setup amortized over three months + $100/month incremental support = $700/month. Against an equivalent $1.09/hour burst GPU, break-even is ~642 hours/month, or 88% of 730 hours. Different CPU/RAM or hardware speed changes that comparison; include power/site costs for delivered equipment. Occasional training rarely reaches this occupancy.

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Operational agents: budget tokens as well as hosts

Plan for roughly five logical roles: campaign supervision, data/knowledge curation, model training/evaluation, quality/follow-up and operations/cost reconciliation. They can share a small agent-runner VM while retaining isolated responsibilities and grants. Many scheduled checks should be deterministic code; an idle agent does not need to call a model.

API pricing anchor	Input / 1m tokens	Output / 1m tokens
GPT-5.4 mini, existing-family reference [S25]	$0.75	$4.50
GPT-5.4, occasional escalation example [S26]	$2.50	$15.00
80% mini / 20% escalation weighted mix	$1.10	$6.60

For 10,000 input and 2,000 billable output tokens/call, the weighted cost is **$0.0242/call**. Budget output/reasoning tokens using actual billed usage, not only visible answer text. The following 50% uplift is a planning allowance for longer outputs, retries and failed turns; tools are additional. Prices are reference-model examples, not a request to change the pinned model or proof that the agents perform their future tasks.

All-agent call rate	Calls / 30 days	Tokens + 50% allowance
Event-driven, 120/day	3,600	$131
Five roles, one call each / 15 minutes	14,400	$523
Five roles, one call each / minute	216,000	$7,841

Base reserve: $750/month total for campaign and operational model/API use. Around $523 covers the middle example, with the balance available for embeddings/tools and campaign reasoning. It is a shared ceiling, not $750 for every agent. Set daily and per-campaign reservation limits; usage can remain far below the ceiling.

Provider tools can cost separately: public OpenAI file-search storage is $0.10/GB/day beyond its free allowance, so a 1-TB hosted corpus would be about $3,000/month before calls. Keep the bulk chemistry corpus in the platform's own artifact/retrieval stores; send scoped context. This report does not assume hosted file-search storage. [S27]

Self-hosting a small model may eventually pay off, but adds GPU idle cost, serving maintenance and quality validation. It should not share an unbounded GPU with deadline-critical training. Choose it from measured cost/quality/privacy requirements, not simply because GPU rental looks cheap.

90-DAY INFRASTRUCTURE PLAN

Choose for continuity, then optimize the worker lane

Default: one primary cloud, CPU-first

AWS is the default recommendation when there is no incumbent ecosystem advantage: the reference bill is directly auditable and the state/artifact/batch components fit the system's boundaries. This is an engineering judgment, not evidence that AWS is universally more reliable or cheaper. Selected Azure CPU prices lower this reference total by about $482/month including reserve; that is meaningful, but smaller than a change in tuning or agent policy.

Choose Azure instead if Microsoft agreements, identity, support or existing operational skill make it easier to run. Choose GCP if the organization already has its data or qualified ML pipeline there. Do not migrate a stable cloud during month two to save a small VM-price difference. No provider has been contracted or authorized for private data disclosure by this estimate.

Starting option	Indicative recurring envelope*	Assessment
One primary cloud; CPU batch; API agents	$5.5–7.5k/month around the base program	Recommended balance: predictable state, elastic training and limited operational surface.
Dedicated CPU hosts + independent backups / optional GPU	$3.5–5.5k/month, quote-dependent	Good if someone owns server operations and local data placement wins. Benchmark memory/throughput and failover before contracting.
Primary cloud + specialist GPU lane	Base bill + measured GPU/storage/egress delta	Useful for actual GPU models. Pre-qualify one supplier, then burst without moving authoritative state.
GPU-first rented cluster / multiple clouds	Often poor fit before GPU demand is known	Adds idle or minimum-bundle cost and operational complexity without helping the v1 CPU paths.

*Ranges are planning assessments, not commercial quotations. The full recurring base/stress arithmetic is on pages 8–9. Engineering labor can reverse the ranking even where bare-metal cash fees are lower.

Separate excluded labor from the infrastructure bill

Illustration only: an extra 2–6 hours/week of server/vendor/network work at $125/hour is ~$1,083–3,250/month. Quote one-time cloud/OS migration, two-campaign support, remote-worker security, reproducible training and restore acceptance separately. Scientific node development, data rights/curation and review are also outside the infrastructure estimate. Do not treat them as included in $6.5k.

Avoid a 1–3-year GPU/server commitment during the first 90 days. A capacity reservation is useful only when the supplier confirms a recurring bottleneck and its cancellation/stock terms fit the plan. A billing discount is not a capacity guarantee.

90-DAY INFRASTRUCTURE PLAN

A 90-day rollout that minimizes infrastructure pivots

Period	Work to complete	Decision / evidence
Before launch	Benchmark selected v1-like models with valid authorized data at 8/16/32 threads; include features, grouped validation, packaging and cold starts. Repeat at forecast data scale. Quote the complete primary-cloud bill.	Fix the primary region, job sizes, training window and cost per accepted release. Verify quota and hardware availability.
Before two-campaign acceptance	Implement/qualify concurrent admission and run-wide caps, fairness, remote tool access, cloud/OS deployment, state migration, backups and restart. Trial both campaigns under training load.	Two campaigns remain an unmet acceptance target until actual full-run and recovery evidence exists. No claim from this budget alone.
Days 1–30	Use on-demand CPU workers, bounded agent/API budgets and immutable artifact storage. Track queue age, job memory, billed hours, accepted releases and provider retries.	Adjust resource sizes or job concurrency inside the chosen deployment. Do not change clouds based on one expensive experiment.
Days 31–60	Right-size from observed distributions; reuse features; remove stale test containers and unused temporary disks. Trial interruptible jobs with accepted checkpoints and fallback.	Savings affect worker usage, not scientific split integrity or the protected campaign baseline.
Days 61–90	Review utilization, spend and node acceptance rate. Negotiate a short reservation or CPU hardware quote only if measured recurring demand justifies it. Rehearse restoration.	Stay on the original platform unless a measured constraint cannot be fixed by capacity or configuration.

Keep portability small and concrete

Use pinned x86_64 Linux packages/images for future workers after qualification, portable model/feature manifests, owned public APIs, checksummed artifacts and checkpointed jobs. Keep provider-specific allocation code behind the Deployment/worker boundary. Do not introduce Kubernetes, a general multi-cloud scheduler or a new graph database solely for two campaigns.

A move from Windows/DPAPI to Linux/cloud secrets requires explicit credential rebinding and recovery qualification. A Linux container label or successful wheel installation is not evidence that native scientific dependencies, storage semantics or owner recovery work on the new host. [L01–L04]

Procurement gate: finish one representative acceptance benchmark and an itemized 90-day quote, then select the primary provider. This is a recommended purchasing decision process; it does not authorize deployment, remote access or paid workload execution.

90-DAY INFRASTRUCTURE PLAN

Pay for recoverability before paying for idle scale

The base bill includes a managed database standby and separate training capacity. It still has a single control-host failure domain. If business availability requires near-continuous service, price a second qualified API/coordinator placement, tested leader fencing and failover—not just a second VM. A live database standby is not an independent backup.

Control	Initial proposal	What remains to verify
Data recovery	Managed SQL point-in-time recovery plus independent versioned artifacts; daily recovery manifest.	Consistency across accepted receipts and artifacts; corrupt/missing objects fail closed; exact dataset/model pins restore.
Recovery objectives	Propose metadata RPO ≤1h and service RTO ≤4h for discussion.	Targets are not current guarantees. End-to-end recovery needs a measured cloud/host failure drill and available staff.
Training failure	Persist immutable inputs and accepted fold/trial checkpoints; return capacity on termination.	No ambiguous model/API retries, stale completion authority or accidental reuse of invalid evaluation output.
Scientific permissions	Scoped worker access, reviewed source rights, supported data location and model disclosure.	Bare-metal tenancy or a Secure Cloud label alone does not settle data contract rights.
Resources / cleanup	Tag owners and job/campaign IDs; kill idle jobs by policy; remove temporary test containers and job disks when finished.	Retain accepted models/history and explicit recovery volumes. Stopping a VM may leave billed disks and IPs.

Meter the things that determine the bill

Record complete worker-hours, per-fit elapsed time, peak resident memory, scratch bytes, dataset upload bytes, model/API billed tokens, retries and cost per scientifically accepted release. Add per-campaign and per-week allocation limits. Report weak or rejected models as failures to qualify; they still consumed compute and must be counted.

Alert at 70% of the monthly provision and escalate at 85%; these are proposed operating thresholds, not implemented controls. Reserve the live campaign and state budget before new exploratory jobs. A limit must stop admitting new discretionary work without corrupting already accepted campaigns or silently changing their pinned policy.

Cross-zone traffic, NAT, registry storage, log retention and persistent GPU volumes require explicit monthly owners. Multi-AZ SQL, object versioning and snapshots can all increase storage beyond the original file size. Reconcile charges against tags rather than assuming all idle resources are free.

90-DAY INFRASTRUCTURE PLAN

The changes most likely to force a budget decision

Change from the base case	Approximate effect	Response within the same platform
20 rather than 10 tuning trials	~26.75 pre-overhead hours/release; CPU training ~$2,944/month instead of ~$1,569.	Measure trial yield; use bounded search/early stopping without weakening the held-out protocol.
Two seeds across every base fold	~101 fits instead of 51; affected training cost nearly doubles.	Require the scientific rationale and price it explicitly before admitting the batch.
Five agents call every minute	~$7,841/month token allowance at the reference prompt/mix, before tools.	Make routine checks deterministic/event-driven; reserve expensive reasoning for meaningful state changes.
250-GB dataset uploaded per release	~$1,950/month outbound at $0.09/GB, plus possible NAT.	Cache dataset revisions at a qualified remote worker supplier or keep training in the primary region.
Large dense features / broad stress tuning	~$17.5k/month CPU training before platform/agents.	Use larger-memory slots, valid feature representations and a new approved spend/throughput envelope.
Deep ML: ~2,167 GPU-hours/month	~$3.9–8.0k/month GPU/VM cost at the compared example rates.	Add the qualified GPU lane. Do not assume the base $147 experiment allowance covers it.

When rental is worth revisiting

Consider dedicated CPU rental when at least four weeks of measured worker occupancy exceeds the quote's full break-even hours, the required RAM is fixed and someone owns replacement/recovery. Consider reserving GPUs when utilization and capacity scarcity are stable, not simply because the calendar month contains a few long runs.

Consider a platform change only if measured availability/region constraints, data-contract requirements, unsupported runtime dependencies or sustained cost beyond the accepted envelope cannot be solved within the selected platform. Growing a worker pool or adding one pre-qualified GPU supplier is not a control-plane migration.

What would make the estimate firmer: actual model/data mix for the first 20 releases; candidate/invoke counts and deadlines for both full campaigns; valid dataset sizes and unique structure counts; tuning/seed/calibration policy; data location/retention obligations; and the required recovery/availability targets.

Confidence: high in separating CPU-first Amphora property training from selected GPU workloads; medium in the reference VM-price arithmetic; low-to-medium in total workload hours until representative benchmarks are run. New science and agents are the largest unmeasured variables.

APPENDIX A / ASSUMPTIONS AND FORMULAS

Calculation basis and audit notes

Quantity	Formula / convention
Planning release count	20 × 52 / 12 = 86.6667/month; 20 × 13 = 260 releases in the 13-week period.
Worker-hours	H = releases × ((trials × folds + final fits) × hours/fit + preprocessing hours) × 1.20.
Base training	86.6667 × ((10 × 5 + 1) × 0.25 + 1.5) × 1.20 = 1,482 VM-hours/month.
Base training cash	1,482 × $1.0584 = $1,568.5488/month. Disk and networking allowances are separate.
Weekly concurrency	ceil(weekly allocated hours / (available window hours × 0.65)); base ceil(342 / 109.2) = 4 slots.
Always-on hours	730h/month for price comparisons; actual provider/calendar usage determines invoices.
Total base provision	(2807.3440 + 1568.5488 + 750 + 146.64) × 1.20 = $6,327.0394.
Agent call cost	(10,000 × $1.10 + 2,000 × $6.60) / 1,000,000 = $0.0242; add stated retry/output allowance and tool charges.
Rental break-even	(monthly quote + setup/3 + incremental operations) / benchmark-adjusted equivalent cloud worker-hour rate.
Data / memory units	Matrix examples use decimal GB. Provider RAM uses GiB where published; GPU vendor memory labels are preserved.

Distinguish two reserves: 20% training overhead is added to training usage; 20% contingency is then added to the full recurring subtotal. Neither is a supplier discount. The $750 API and $147 GPU lines are bounded allowances, not predicted mandatory consumption. Their inclusion makes the provision conservative.

Price observation: 2026-10-14. AWS public feed publication 2026-10-12 19:39:25 UTC; Linux, US East (N. Virginia), on-demand. Azure query selects East US, Consumption and named VM SKUs, excluding Windows, Spot, Low Priority and similarly named Cloud Services meters. GCP public tables were fetched and read directly; region/stock must be confirmed before a quote. No ARM worker compatibility is assumed. [S01] [S03] [S04] [S05] [S06]

No false equivalence: prices exclude tax; they are not performance-normalized. T4 16 GB, L4 24 GB, A100 40/80 GB and H100 differ. GPU price/GPU-hour, full multi-GPU node-hour, CPU-vCPU-hour and complete VM-hour must not be interchanged. Verify CPU/RAM, storage, commitment and region before provisioning.

APPENDIX B / PRIMARY SOURCES / CHECKED 14 OCTOBER 2026

Sources: cloud infrastructure

References are clickable. Prices are public observations or directly fetched price-feed records, not negotiated quotes. Date-sensitive provisioning checks remain required.

S01 — AWS EC2 public Linux price feed, US East (N. Virginia)

S02 — AWS EC2 on-demand billing and transfer pricing

S03 — Microsoft Azure Retail Prices API documentation

S04 — Azure public retail prices: East US Linux VM query

S05 — Google Compute Engine general-purpose VM pricing

S06 — Google accelerator-optimized VM pricing

S19 — AWS Batch pricing

S20 — Azure Batch costs

S21 — Google Batch pricing

S22 — AWS Spot interruption semantics

S23 — Azure Spot eviction / SLA semantics

S24 — Google GPU / Spot guidance

S28 — AWS VPC / NAT pricing

S29 — AWS S3 gateway endpoints

S32 — AWS RDS for PostgreSQL pricing scope

S01 is an AWS-hosted public price-page feed, retrieved without credentials; its publication timestamp is recorded on page 19. S04 is the actual Microsoft retail query used, with the filtering distinctions above. GCP HTML was fetched directly because the browsing renderer could not open the very large pricing tables.

APPENDIX B / PRIMARY SOURCES / CHECKED 14 OCTOBER 2026

Sources: specialist compute, rental and APIs

S07 — Runpod public GPU and storage pricing

S08 — Runpod cloud tiers and disk persistence

S09 — Lambda: single- and multi-GPU instance pricing

S10 — Lambda: billing and filesystem charges

S11 — Hyperstack GPU and storage pricing

S12 — CoreWeave public instance pricing

S13 — Paperspace / DigitalOcean public pricing

S14 — Vast.ai official pricing guide

S15 — OVHcloud dedicated GPU server catalog

S16 — Hetzner AX102 CPU server specifications

S17 — OVHcloud US Advance bare-metal server catalog

S18 — Rentacomputer: server / GPU rental quotation

S25 — OpenAI GPT-5.4 mini pricing

S26 — OpenAI GPT-5.4 pricing

S27 — OpenAI API tools and storage pricing

S30 — scikit-learn parallelism and oversubscription

S31 — LightGBM GPU tuning guidance

Runpod list values require tier/region confirmation; Paperspace headline prices have product conditions; CoreWeave training bundle examples are whole-node charges. No marketplace “lowest offer,” introductory credit or temporary promotion is used to make the reference platform fit. Rental alternatives need written configuration/setup/support quotes.

APPENDIX C / REPOSITORY PROVENANCE

Sources: local implementation and model evidence

L01 — Current deployment and one-campaign boundary
v2/docs/Architecture/System/Runtime_Topology.md; v2/docs/Architecture/Requirements/Product_Requirements.md

L02 — Accepted deployment limits and successor triggers
v2/docs/Audits/BP-09_Selected_Profile_Handoff.md

L03 — Active population admission contract gap
v2/docs/Build Plans/POP-01.01 Population Execution.md; POP-01.05 Bounded graph admission decision.md

L04 — Inherited model integration and scope limits
v2/docs/Build Plans/ML-01 Inherited Scientific Model Integration.md; v2/docs/Build Plans/Master Build Tracker.md

L05 — Later BigSolDB primary-protocol workload and timing
v1/amphora-solvency-score/benchmarks/bigsoldb-v1/inhouse_benchmark_receipt.v1.dsse.json; run_v20_inhouse.py; run_benchmark.py; featurize.py

L06 — Earlier solubility model workload receipt
v1/amphora-solvency-score/reference/mvp20/data/bigsoldb_solubility_model_receipt.v1.json

L07 — Polyverse viscosity workload receipt
v1/amphora-viscosity-score/reference/mvp20/data/polyverse_melt_viscosity_model_receipt.v1.json

L08 — Polyverse diffusivity workload receipt
v1/amphora-barrier-corrosion-score/reference/mvp20/data/polyverse_solvent_diffusivity_model_receipt.v1.json

L09 — Purformance historical workload receipts
v1/amphora-design-rule-score/reference/purformance/zenodo-13742098/purformance_multi_target_coating_model_receipt.v1.json; purformance_scratch_hardness_model_receipt.v1.json

L10 — Separate NOMAD GPU program and feature-bank boundary
v1/README.md (Layout); v1/CRYSTALATLAS.md; v1/nomad_crystal360/models.py; atlas_worker.py; v1/run_atlas_screen.py

L11 — Additional classical model families
v1/amphora-solvency-score/reference/mvp20/data/polyomics_dmf_chi_model_receipt.v1.json; v1/amphora-molecule-property/benchmarks/phys-ml-stack-comparison-c8-v1/c8_compute.py; v1/amphora-molecule-role-ranker/benchmarks/role-parsimony-conformal-v1/run_role_parsimony_conformal.py

Local documents are implementation/provenance references, not externally published source links. Their historical scientific metrics were not revalidated. The v1 tree was read only: no training, formatting, builds or operational automation ran there. No datasets or credentials are embedded in this PDF.

Report status: business planning estimate. Two-campaign production capacity, model qualification and provider/host acceptance require the separate delivery evidence identified in this report.